

**Full Name:** \_\_\_\_\_  
**EEL 3135 – Classwork #08**

**Milestone:**

- Complete any 2 questions to receive credit.

**Question #1:** Compute the following by hand:

$$y[n] = (2\delta[n+1] - \delta[n-2] + 3\delta[n-4]) * (\delta[n-1] + 2\delta[n-3])$$

- (a) Write out all shifted and weighted impulse terms explicitly.
- (b) Simplify to a single expression for  $y[n]$  using discrete convolution properties.
- (c) Plot the resulting  $y[n]$  from part (2) in MATLAB for  $n = -5:10$  using the `stem()` command. Label all nonzero impulses clearly and include proper axis titles.

**Question #2:** Two FIR systems are connected in cascade:

$$h_1[n] = \frac{1}{2}(\delta[n] + \delta[n-1]), \quad h_2[n] = \delta[n] - \delta[n-2]$$

with input  $x[n] = u[n]$ .

(a) Compute the overall impulse response  $h_T[n] = h_1[n] * h_2[n]$ .

(b) Determine the system output  $y[n] = x[n] * h_T[n]$ .

(c) Is the overall system **causal** and **FIR**? Justify your answer.

**Question #3:** *Given:* Let  $x = [1, -1, 2, 0]$  and  $h = [2, 1]$ .

(a) Write MATLAB code that constructs the **Toeplitz matrix** for  $x$  and computes  $y = T_x * h(:)$ .

(b) Compare the result to `conv(x, h)` and confirm they match.

(c) Briefly explain why both methods yield identical outputs according to the *commutative property* of convolution.

**Question #4:** *Given:* A discrete-time signal is given as:

$$x[n] = \sin(0.2\pi n) + 0.5w[n],$$

where  $w[n]$  is white Gaussian noise. A length-5 moving average FIR filter is applied:

$$h[n] = \frac{1}{5}(u[n] - u[n - 5])$$

(a) Write MATLAB code to generate  $x[n]$ , apply the filter using `conv(x, h, 'same')`, and plot both signals.

(b) Estimate qualitatively how the filter affects the frequency content (i.e., explain why it smooths  $x[n]$ ).

(c) If the filter length is increased to 20, describe the trade-off between **noise reduction** and **signal delay**.